

# Lung Cancer: An interplay between Behavioural & Psychosocial Factors with Early Clinical Markers

Isaac Chukwuemke Oselukwue  
Research Methods Module

Master of Computer Science (In View)  
University of Lincoln

Lincoln, England  
29353479@students.lincoln.ac.uk

**Abstract**—Lung Cancer is till date a terrifying disease with very low survival rates. This report carries out statistical analysis exploring interactions between behavioural and psychosocial interactions with Lung Cancer clinical markers. Data preprocessing and descriptive statistics was applied on dataset. Pearson's correlation was applied and displayed ALCOHOL\_CONSUMING ( $r = 0.2944$ ) and ALLERGY ( $r = 0.3336$ ) had close correlation with Lung cancer. Further analysis using inferential statistics was applied on 3 models: Base Model, Interaction Model and Full Model with likelihood ratio chose Interaction Model as best model ( $\chi^2 = 13.24$ ,  $p = 0.0102$ ) and saw strong interactions on the behavioural and psychosocial factors with lung cancer with ALCOHOL\_CONSUMING, ALCOHOL\_ANXIETY and PEER\_PRESSURE leading. Anova test was carried out on different age groups categorising them by risk profiles, gender and behavioural and psychosocial markers supporting Risk\_Level and Smoking\_Alcohol influenced lung cancer. Bayesian statistics using Watanabe–Akaike information criterion (WAIC) and Leave-one-out cross-validation (LOO-CV) with results from WAIC supporting base model with Elpd of  $-92.70$  while LOO-CV also supported Interaction model as best model with Elpd of  $182.96$ .

## I. INTRODUCTION

Lung cancer, a killer carcinoma is the most frequent cause of cancer deaths experienced amongst cancer patients globally; with tracking rate being usually tedious to compile due to data collation and quality implications, has seen its mortality rate stay relatively constant as a major killer having a 28% rate in the 2010s [1] and a decade later 22% in 2020s [2]. This form of cancer is often attributed to smokers [3], with traces of being dated to the massive marketing of cigarettes before World War II [8], however in recent reports, 1 in 4 lung cancer patients have never smoked before [4]. Lung cancer could also be contracted through exposure to harmful air pollutants [9], type 2 diabetes [10] and even hereditary [11]. Lung cancer severity is broken into stages from 0 to IV with IV being the most severe form of it [5]. Many efforts are being invested to detect it early enough especially at stage 0 from wedge resection with a high recovery rate [5]; in stage I, a more invasive lobotomy is required to remove the affected lung lobe; in stage IV often referred to as the Metastatic stage, treatment is usually very painful and mainly experimental especially stage IVB. What is even more worrisome is the post survival rate after a successful treatment with only 76% surviving the first 24 months post treatment [6] and an even lower 30% do not make it past 5 years [7].

The psychological struggles of which include depression, anxiety and social isolation is a key characteristic observed in lung cancer patients with over 3 in 10 patients [12] experiencing and exhibiting these struggles. The factors that typically trigger these have been linked to long stays in hospitals receiving treatment, loss of strength in carrying out physical activities and abandonment from friends during the treatment periods [13]. Analysis on typical behavior of lung cancer patients revealed that 99 in 100 Chinese nationals started smoking due to peer pressure [14]. Behavioral patterns such as smoking, alcohol consumption does not seem to slow

down even after patients have been informed of their diagnosis with about 4% of patients still continuing with smoking and 18% of peers still smoking; also 43% of patients still continued alcohol consumption and over 71% of their family members doing so as well [15].

The interplay between lung cancer and psychosocial and behavioural factors was supported by the UK Biobank which carried out research concluding that without the smoking variable, psychological distress increased lung cancer risks [16]. Even post recovery, only 8 in 100 patients were living healthily and meeting properly planned food requirements.

## II. LITERATURE REVIEW

### A. Behavioural Risk Factors

In a study done comparing colorectal cancer to lung cancer, 78 in 100 lung cancer patients were obese and the higher the age of the individual, the lower the low muscle mass with a 75% likelihood [17]. The average age for lung cancer is between 40 to 80 years old with median of 63 years old [18]. The likelihood of death increases after the age of 60 for patients with low fruit diet and 80 for smokers [19].

Between the 1990s and 2021, there is an overall increase in the death rate for lung cancer with a percent change of 62.05% for smokers with women having a higher increase of 88% and men 57% [19]. Further research was carried out to investigate the relationship of smoking to bodily functions in lung cancer patients, it was observed that an abnormally high metabolism rate and stronger connectivity is present in the individuals [20]. Due to insufficient healthcare infrastructure and workers, proposals to deprioritise smokers (people with option luck) from health treatments are being proposed [21] to promote fairness. In a study comparing behavioral habits such as smoking, alcohol consumption, physical inactivity and similar lifestyle creeps for 16,000, 48% of them performed these habits have a strong influence leading to lung cancer [24].

Dietary patterns have been proven to influence lung cancer inversely with studies showing that people consuming fish, red meat, fresh fruits and vegetables had a lower risk of having lung cancer while processed meat such as bacon increased the likelihood of lung cancer [23]. Further studies on natural remedies like daily consumption of green tea, reduced odds of lung cancer by 8.2% [25]. Quite a number of reports have come out in the past evidencing the association of vitamin b12 with lung cancer. However, more recent studies seem to be inconclusive [26], suggesting no direct relationship between them when taken in an optimal range.

### B. Psychological and Sociological Burden

With 25% of lung cancer patients experiencing stigma and increased depression blame themselves and regret many actions [22], the psychological and social factors are quite key areas to be studied.

Anxiety and depression, common stressors are typically exhibited by lung cancer patients and in comparison, to other

terminal diseases such as covid-19 [27]. Even in post-treatment, anxiety biomarkers are still very present in recovering patients scaring them with PTSDs that affect cognition and lowers quality of life [28]. Research also further suggests that these stress related burdens are as a result from withdrawal from ever smokers [29].

Survivors who have lived the longest have been supported emotionally and socially by peers and family and this enabled them pull through [30].

### C. Research Question

Exploring various papers, there are none that explore the interactions between the factors such as anxiety and peer pressure which are under psychosocial and other factors like smoking and alcohol consumption which are under behavioral habits especially across age and gender demographics. With this in mind I propose the null hypothesis to be:

$H_0$  (null hypothesis): *There is no significant interaction; the impact of psychosocial and behavioral habits on lung cancer is the sum of their independent events and no integration exists.*

I also propose the alternative hypothesis to be:

$H_1$  (alternative hypothesis): *There is a significant interaction; implying that individuals that experience psychosocial stress and behavioral habits have a high risk of lung cancer and this integration is moderated by age and gender.*

My primary research question aims to answer: whether the interplay between psychosocial factors and behavioral habits increases likelihood of cancer and if it is moderated by age and gender.

## III. RESEARCH METHODS

### A. Dataset Description

The dataset used for this experiment was obtained from Kaggle with cc0 license for the dataset. The dataset has been vetted for ethical compliance and no markers for patients identification is in the dataset. The dataset consists of 309 records and 16 distinct features which are compiled to address research in lung cancer patients. Data pruning and purging was carried out on the dataset to remove duplicates, applying binary categorisation on categorical variables which reduced the dataset to 276 entries. The dataset had mainly categorical variables. The only continuous variable in this dataset was age. Software used to carry out analysis is Python 3.12 in Jupyter notebook using statical libraries (Pymc, pandas, numpy) with GitHub repository in appendix.

### B. Descriptive Statistics

The variable ‘age’ has a relatively symmetrical distribution with a mean of 62.91 years and median of 62.50 years. The age data varied from 21 to 87 years. The standard deviation (SD) of 8.38 and variance of 70.21 indicates moderate variability of the dataset. 50% of individuals fall within 11.25-year range due to the first quartile (Q1) of 57.75 and third quartile (Q3) of 69.00.

The categorical variables had a healthy mix of patients’ characteristics. The gender distribution is nearly balanced with more male participants comprising 51.45% compared to females for 48.55%. A modest skew was also observed for behavioural factors: smokers – 54.35%, alcohol – 55.07%.

Clinical markers such as yellow fingers, coughing, shortness of breath is observed in more than 50% of participants. Finally, the outcome ‘Lung Cancer’ is present having a substantial majority of 86.23%. The primary categorical variables can be found below.

TABLE I. PRIMARY CATEGORICAL VARIABLES

| S/ N | Variable          | Unique Values | Value | Frequency | Percentage (%) |
|------|-------------------|---------------|-------|-----------|----------------|
| 1    | SMOKING           | 2             | 1/0   | 150/126   | 54.35/45.65    |
| 2    | ANXIETY           | 2             | 1/0   | 137/139   | 49.64/50.36    |
| 3    | PEER_PRESSURE     | 2             | 1/0   | 140/136   | 50.72/49.28    |
| 4    | ALCOHOL_CONSUMING | 2             | 1/0   | 152/124   | 55.07/44.93    |
| 5    | LUNG_CANCER       | 2             | 1/0   | 238/38    | 86.23/13.77    |

Looking at the distribution by individual features, the behavioural and psychosocial variables indicate some interesting patterns for this dataset. 84.02% of smokers and 87.33% of never smokers have lung cancer while other variables have larger differences such as individuals with anxiety, yellow fingers and peer pressure have lung cancer rates > 90% while the absence is much lower.

Also, looking into the interplay between psychosocial factors: anxiety and peer pressure, respondents with Yes\_Yes have a high lung cancer rate of 96.43%. Same experience is noted for behavioural factors: smoke and alcohol consumption (Yes\_Yes) have as well a high rate of cancer at 96.2%. Respondents with none of them had a much lower rate at 71.7%.

### C. Correlation Analysis

An initial analysis was carried out on the dataset using Pearson and Spearman’s Correlation to get the top correlations among the variables. The top observed correlation was between ANXIETY and YELLOW\_FINGERS variables with  $r = 0.5583$  which indicates a strong link between these 2.

TABLE II. TOP 3 CORRELATIONS

| Variable A            | Variable B     | Correlation |
|-----------------------|----------------|-------------|
| ANXIETY               | YELLOW_FINGERS | 0.5583      |
| SWALLOWING_DIFFICULTY | ANXIETY        | 0.4788      |
| ALCOHOL_CONSUMING     | GENDER         | 0.4343      |

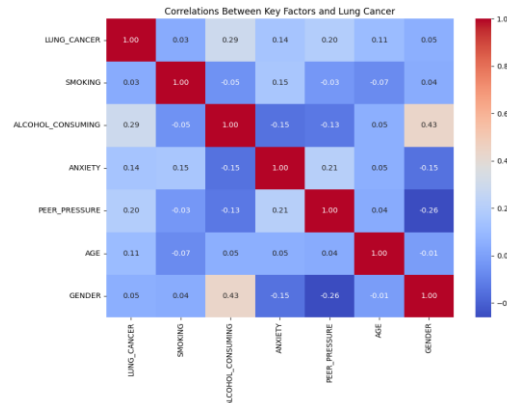


Fig. 1: Correlation between Key Factors and Lung Cancer

On direct correlations with Lung Cancer, ALCOHOL\_CONSUMING ( $r = 0.2944$ ) and ALLERGY ( $r = 0.3336$ ) display a moderate positive correlation with lung cancer. The psychosocial factors ANXIETY (0.1443) and PEER\_PRESSURE 0.1951 diverge with lower correlations which indicates that for this dataset, behaviour factors are greater associated with lung cancer.

#### IV. RESULTS

Three models were explored for both Inferential and Bayesian statistics. The Base Model for just the main effects, the Interaction Model that adds the Psychosocial and Behavioural Interactions and the Full Model adding age and gender.

##### A. Inferential Statistics Using Logistic Regression

Three regression models were fitted; I used likelihood ratio tests and it indicated that the addition of psychosocial and behavioural factors (Interaction Model) improved the model to produce a much better fit ( $\chi^2 = 11.05$ ,  $df = 8$ ,  $p = 0.1991$ ) compared to the base model ( $\chi^2 = 13.24$ ,  $df = 4$ ,  $p = 0.0102$ ). The Interaction model has a log-likelihood = -79.17, AIC = 180.35, BIC = 220.17,  $R^2 = 0.2842$  showed ALCOHOL\_CONSUMING has a strong positive association ( $\beta = 4.1713$ ; Odds Ratio = approximately 64.8;  $p < 0.001$ ); interaction effects: ALCOHOL\_ANXIETY ( $\beta = -2.2142$ ; OR = 0.1092;  $p = 0.0235$ ) and PEER\_PRESSURE ( $\beta = 1.4326$ ; OR = 4.1894;  $p = 0.0268$ ) were significant. Model performance had a classification accuracy = 86%, AUC = 0.87.

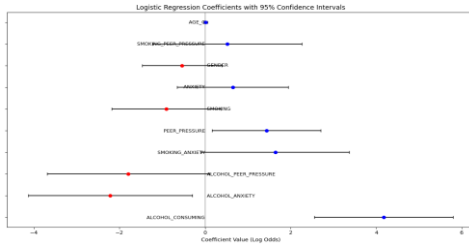


Fig. 2: Logistic Regression Coefficient Plot

##### 1) Inferential Statistics Using Confidence Interval (CI)

Comparing the Base with the Interaction Model, taking ALCOHOL\_CONSUMING displays a coefficient ( $\beta$ ) = 4.1713, 95% CI = (2.5477, 5.7949), corresponding odds ratio OR = (12.78, 328.63) and 65x more likely. ALCOHOL\_ANXIETY  $\beta = -2.2142$ , CI = (-4.1307, -0.2977), OR CI = (0.0161, 0.7425) indicating impact of alcohol consumption is moderated significantly by anxiety. The PEER\_PRESSURE  $\beta = 1.4326$ , CI: (0.1647, 2.7004), OR CI: (1.1791, 14.8856) with 4x more likely supports the insights that psychosocial factors are interacting with behavioural factors to increase risk of lung cancer.

Other main effect variables: GENDER, SMOKING, ANXIETY, and AGE\_C, the CI include 0 (or OR include 1) which implies that these variables are not significantly affecting lung cancer risk for this model.

##### 2) Inferential Statistics Using Anova Test

To further look into group-level differences, I introduced 3 Risk Levels based on lung cancer rates using KMeans clustering on the dataset to standardise it with a random state of 42 which has an [F (2, 273) = 6.2666,  $p = 0.0022$ ] with

further validation using Tukey's HSD tests showed there was a major difference in High and Low Risk groups (mean difference = 0.1281,  $p = 0.0342$ ), comparing by GENDER, the difference was not much [F (1, 274) = 0.7914,  $p = 0.3745$ ]. Another validation on behavioural patterns with groups: 'Alcohol Only', 'Both', 'Neither', 'Smoking Only' was quite significant [F (3, 272) = 8.9771,  $p < 0.001$ ]. Comparing 'Alcohol Only' and 'Neither' showed they differ significantly, 'Both' and 'Smoking Only' also differ significantly meaning lung cancer is directly influenced by these pairs.

##### B. Bayesian Analysis using Bayesian Logistic Regression

Bayesian Logistic Regression was applied on the same 3 models with 4 chains, 2000 draws and 5000 tunes. Using Widely Applicable Information Criterion (WAIC) the base model was preferred with a score = -92.57. I observed the difference between base model and the others were less than 1 which prompted me to apply Leave-One-Out Cross-Validation (LOO-CV) to validate the results due to LOO-CV being preferred because of the standard estimate error [31]. LOO-CV supported the observation from Logistic Regression and favours the Interaction model (rank = 0, elpd\_loo=182.96, ploo = 7.079819) suggesting interactions improved performance. The standard error gotten from the difference between Interaction and Full = 0.8 is low suggesting no meaningful difference between both of them.

TABLE III. MODELS EVALUATION WITH LOO-CV

| Model       | Rank | Elpd_loo | P_loo | Dse  |
|-------------|------|----------|-------|------|
| Interaction | 0    | 182.96   | 7.07  | 0.00 |
| Full        | 1    | 183.77   | 9.58  | 1.96 |
| Base        | 2    | 185.19   | 6.02  | 1.76 |

Applying Approximate Bayes Factor had a Bayes Factor of 13,645 comparing Interaction with Bayes and an extremely high value comparing interaction with Full Model indicating the full model has no predictive gain beyond the interactions alone. Interaction Model performance had AUC = 0.8224 and classification accuracy = 87%.

##### 1) Bayesian Analysis with 95% Credible Intervals

ALCOHOL\_CONSUMING Mean coefficient 2.0248 (95% CI: [1.1925, 2.8921]); odds ratio 8.33 (95% CI: [3.30, 18.03]), PEER\_PRESSURE (Interaction Effect) Mean coefficient 1.1733 (95% CI: [0.4092, 1.9606]); odds ratio 3.50 (95% CI: [1.51, 7.10]) showing both a strong effect and significant contribution. ANXIETY, AGE\_C, SMOKING, and GENDER did not show significant effects.

##### 2) Bayesian Analysis using MCMC Trace Diagnostics

Trace plots for SMOKING, ALCOHOL\_CONSUMING, ANXIETY, PEER\_PRESSURE, SMOKING\_ANXIETY showing kernel density estimates for multiple chains and converge to a unimodal distribution showing it is stable. The progression over iterations mixes well and bounded by a common mean and reached stationary distribution.

#### V. CONCLUSION AND FUTURE WORKS

This paper explored Lung cancer and its interplay between Psychosocial and Behavioural Factors interaction with Lung Cancer Markers. Collectively, likelihood ratio test, Anova test, Bayes Regression upheld the alternative hypothesis that lung cancer is not just the sum of independent psychosocial

and behavioural factors. The Bayesian Analysis showed that factors such as alcohol consumption and peer pressure increase the risk of lung cancer substantially with odds ratio exceeding 3 and 10 respectively. The likelihood ratio test confirmed this with  $\chi^2 = 13.24$ ,  $p = 0.0102$  identifying similar factors as well as addition of ALCOHOL\_ANXIETY. The ANOVA results also supported the evidence with significant differences detected in lung cancer by Risk\_Level and Smoking\_Alcohol supported that those with high peer pressure or those who smoke and consume alcohol have clear lung cancer risks in comparison with others. The analyses all reject the null hypothesis ( $H_0$ ) that the impact of psychosocial and behavioral factors on lung cancer operates independently.

This study is however limited to only 276 patients which implies unforeseen biases based on geography, clinical stage or societal status. Further qualitative works would be recommended in the future aiming a balance of patients from high and low risk clusters with variation in mind. Also, selected interviews on why peer pressure is influential and how anxiety is moderating alcohol. These would help with better recommendations to implement.

#### REFERENCES

- [1] K. Gately, \*Lung Cancer: A Comprehensive Overview\*. New York: Nova Science Publishers, Inc., 2013, pp. 2-16, 20. (Cancer Etiology, Diagnosis and Treatments).
- [2] World Health Organisation, "Cancer," 2025. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/cancer>.
- [3] International Association for the Study of Lung Cancer, "IASLC Language Guide," 2021.
- [4] J. LoPiccolo, A. Gusev, D. Christiani, P. Jänne, "Lung cancer in patients who have never smoked — an emerging disease," \*Nature Reviews Clinical Oncology\*, vol. 21, no. 2, pp. 121–146, 2024, doi: 10.1038/s41571-023-00844-0.
- [5] S. Sandeep, K. Ramakrishnan, and K. Murali, "AI-based detection of different stages of lung cancer using MRI/CT imaging," in \*Proc. 2024 First Int. Conf. Innov. Commun., Electr. Comput. Eng. (ICICEC)\*, Davangere, India, 2024, pp. 1–6, doi: 10.1109/ICICEC62498.2024.10808961.
- [6] A. Ghazipour \*et al.\* , "Survival outcome prediction for stereotactic body radiation therapy of lung cancer from post-RT Ct images with RNN/CNN deep learning," in \*Proc. 2023 IEEE 20th Int. Symp. Biomed. Imaging (ISBI)\*, Cartagena, Colombia, 2023, pp. 1–4, doi: 10.1109/ISBI53787.2023.10230647.
- [7] F. Tichanek, A. Försti, O. Hemminki, A. Hemminki, and K. Hemminki, "Survival in lung cancer in the Nordic countries through a half century," \*Clinical Epidemiology\*, vol. 15, pp. 503–510, May 2023, doi: 10.2147/CLEP.S406606.
- [8] D. R. Shopland, "Tobacco use and its contribution to early cancer mortality with a special emphasis on cigarette smoking," \*Environmental Health Perspectives\*, vol. 103, no. 8, pp. 131–142, 1995, doi: 10.1289/ehp.95103s8131.
- [9] M. C. Turner \*et al.\* , "Outdoor air pollution and cancer: an overview of the current evidence and public health recommendations," \*CA: A Cancer Journal for Clinicians\*, vol. 70, no. 6, pp. 460–479, 2020, doi: 10.3322/caac.21632.
- [10] C. Frank, M. Fallah, J. Ji, J. Sundquist, and K. Hemminki, "The population impact of familial cancer, a major cause of cancer," \*International Journal of Cancer\*, vol. 134, pp. 1899–1906, 2014, doi: 10.1002/ijc.28510.
- [11] X. Liu \*et al.\* , "Cancer risk in patients with type 2 diabetes mellitus and their relatives," \*International Journal of Cancer\*, vol. 137, pp. 903–910, 2015, doi: 10.1002/ijc.29440.
- [12] J. Lynch, F. Goodhart, Y. Saunders and S. J. O'Connor, "Screening for psychological distress in patients with lung cancer: results of a clinical audit evaluating the use of the patient Distress Thermometer," \*Supportive Care in Cancer\*, pp. 193–202, Jan. 2010.
- [13] X. Yan, X. Chen, M. Li and P. Zhang, "Prevalence and risk factors of anxiety and depression in Chinese patients with lung cancer: a cross-sectional study," \*Cancer Management and Research\*, vol. 11, pp. 4347–4356, 2019.
- [14] L. Zhang, J. F. Ye, and X. Zhao, "'I saw it incidentally but frequently': Exploring the effects of online health information scanning on lung cancer screening behaviors among Chinese smokers," \*Health Communication\*, vol. 40, no. 2, pp. 345–356, 2025, doi: 10.1080/10410236.2024.2345948.
- [15] M. E. Cooley \*et al.\* , "Health behaviors, readiness to change, and interest in health promotion programs among smokers with lung cancer and their family members: a pilot study," \*Cancer Nursing\*, vol. 36, no. 2, pp. 145–154, Mar.–Apr. 2013, doi: 10.1097/NCC.0b013e31825e4359.
- [16] NHS Healthy London Partnership, "Improving psychologically informed cancer care: implementing the London Integrated Cancer Psychosocial Care Pathway and the development of psycho-oncology services: A business case," Feb. 2020.
- [17] T. Abbass \*et al.\* , "CT derived measurement of body composition: Observations from a comparative analysis of patients with colorectal and lung cancer," \*Nutrition & Cancer\*, vol. 77, no. 1, pp. 70–78, 2025, doi: 10.1080/01635581.2024.2392913.
- [18] L. Tian \*et al.\* , "Association between sarcopenia, clinical outcomes, and survival in patients with extensive-stage small cell lung cancer treated with first-line immunochemotherapy: A prospective cohort study," \*Nutrition & Cancer\*, vol. 77, no. 1, pp. 62–69, 2025.
- [19] L. Li \*et al.\* , "Disease burden of lung cancer attributable to metabolic and behavioral risks in China and globally from 1990 to 2021," \*BMC Public Health\*, vol. 25, no. 1, pp. 1–16, 2025.
- [20] H. Wang \*et al.\* , "Organs and systems glucose metabolism analysis in different smoking groups for lung cancer patients using total-body PET/CT," in \*Proc. 2024 IEEE Nucl. Sci. Symp. Med. Imaging Conf. Room Temp. Semicond. Detect. Conf. (NSS/MIC/RTSD)\*, Tampa, FL, USA, 2024, p. 1, doi: 10.1109/NSS/MIC/RTSD57108.2024.10656265.
- [21] G. Vijapurkar, "Healthcare resource allocation: Smoking, lung cancer, and the National Health Service," \*World Medical Journal\*, vol. 70, no. 2, pp. 46–48, 2024.
- [22] K. Chansky, M. Rigney, and J. C. King, "Real-world analysis of the relationships between smoking, lung cancer stigma, and emotional functioning," \*Cancer Medicine\*, vol. 13, pp. e6702, 2024, doi: 10.1002/cam4.6702.
- [23] W.-Y. Lim \*et al.\* , "Meat consumption and risk of lung cancer among never-smoking women," \*Nutrition & Cancer\*, vol. 63, no. 6, pp. 850–859, 2011.
- [24] E. Roos \*et al.\* , "Pairwise association of key lifestyle factors and risk of solid cancers - A prospective pooled multi-cohort register study," \*Preventive Medicine Reports\*, vol. 38, pp. 102607, 2024, doi: 10.1016/j.pmedr.2024.102607.
- [25] C.-M. Oh \*et al.\* , "Consuming green tea at least twice each day is associated with reduced odds of chronic obstructive lung disease in middle-aged and older Korean adults," \*Journal of Nutrition\*, vol. 148, no. 1, pp. 70–76, 2018.
- [26] N. T. Le \*et al.\* , "Vitamin B12 intake and cancer risk: Findings from a case-control study in Vietnam," \*Nutrition & Cancer\*, vol. 77, no. 2, pp. 252–264, 2025.
- [27] T. R. Blevins \*et al.\* , "COVID-19 or cancer stress? Anxiety and depressive symptoms in patients with advanced lung cancer," \*International Journal of Behavioral Medicine\*, vol. 31, no. 2, pp. 325–330, 2024.
- [28] J.-R. Wu, V. C.-H. Chen, Y.-H. Fang, C.-C. Hsieh, and S.-I. Wu, "The associates of anxiety among lung cancer patients: Dehydroepiandrosterone (DHEA) as a potential biomarker," \*BMC Cancer\*, vol. 24, no. 1, 2024, doi: 10.1186/s12885-024-12195-9.
- [29] C. Bozkurt and H. Bulut, "The association of depression, anxiety, stress, and quality of life with the presence or absence of chronic obstructive pulmonary disease in lung cancer patients," \*Applied Nursing Research\*, vol. 83, 2025, doi: 10.1016/j.apnr.2025.151933.
- [30] J. Heo and Y. Kim, "A Comparative Study of Symptoms, social support, and quality of life at different survival stages of lung cancer patients," \*Journal of Korean Academy of Fundamentals of Nursing\*, vol. 32, no. 1, pp. 128–137, 2025.
- [31] A. Vehtari, G. Andrew and J. Gabry, "Practical Bayesian Model Evaluation Using Leave-One-out Cross-Validation and WAIC," \*Statistics and Computing\*, vol. 27, no. 5, pp. 1413-1432, 2017.